Name: _____

THE MULUNGUZI CLUSTER EXAMINATIONS BOARD

2018 MALAWI SCHOOL CERTIFICATE OF EDUCATION MOCK EXAMINATION

AGRICULTURE

Subject Number: M012/I

Tuesday March 20, 2018

Time Allowed: 2 hours

08:00 – 10:00 am

PAPER I (100 marks)

Instructions:

1. **This paper contains 10 pages. Please check.**
2. Write your **name** at the top of **each** page of the question paper.
3. This paper has **two** sections: **A** and **B**
4. Section A has **ten** questions and section **B** had **four** questions.
5. Answers all questions in the spaces provided.
6. In the table provided on the page, tick against the questions numbers you have answered.
7. At the end of the examination, hand in your paper to the invigilator.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
Total			

PLEASE DO NOT TURN OVER UNTIL YOU ARE TOLD TO DO SO.

Section A (60 Marks)

Answer **all ten** questions in this section.

1.

a. List any **two** types of soil structure.

_____

_____

_____(2 marks)

b.

i. Give any **three** ways in which soil structure can be improved.

_____

_____

_____

_____

_____(3 marks)

ii. Explain **one** importance of Caution Exchange Capacity on plant growth.

_____

_____

_____

_____(2 marks)

2.

a. Give **three** characteristics of livestock to be selected for breeding

_____

_____

_____

_____

_____(3 marks)

b.

i. Explain **one** limitation of inbreeding.

_____

_____(2 marks)

ii. State any **two** advantages of artificial insemination.

_____

_____

_____

_____(2 marks)

iii. Distinguish between progeny testing and sib selection.

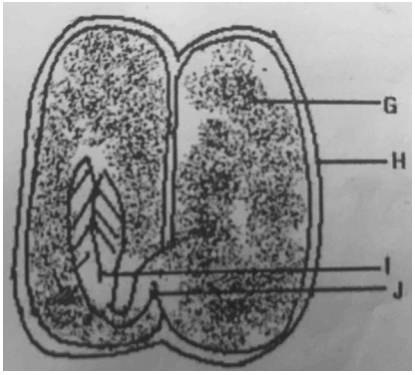
_____

_____

_____

_____(2 marks)

3. Study the figure below of a plant seed and answer the questions that follow.



a. Identify the seed type.

_____(1 mark)

b. Name the parts labeled G, H, I and J.

G:_____

H: _____

I: _____

J: _____(2 marks)

c. What are the functions of the parts labeled G and J?

G:_____

J:_____(2 marks)

d. Give **two** main reasons that make it necessary to use the vegetative parts of a plant to propagate crops.

_____

_____

_____

_____(2 marks)

4. Table 1 shows a wrong randomized block design of an experiment on maize variety trial by students at Phiria secondary school. Use it to answer the questions that follow.

	BLOCK 1	BLOCK 2	BLOCK 3
PLOT 1	NSCM 41	MH 16	NSCM 41
PLOT 2	Local Maize	MH 12	Local Maize
PLOT 3	MH 16	Local Maize	MH 12
PLOT 4	MH 12	MH 15	NSCM 41

Table 1

a. Mention **two** mistakes in the experimental design

_____

_____

_____

_____(2 marks)

b. Apart from randomization, explain **two** other ways in which students could have made the results of the experiment more reliable.

_____

_____

_____

_____(2 marks)

c. Explain any **two** ways in which the results from this experiment could have been analyzed.

_____

_____

_____(2 marks)

5.

a. State any **two** agricultural technologies which ensure food security.

_____

_____

_____

_____(2 marks)

b. Give **one** reason why cassava ensures food security for the growing population.

_____

_____

_____(1 mark)

c. Explain **two** ways in which proper food storage ensures food security.

_____

_____

_____

_____(2 marks)

6.

a. Explain any **two** ways in which pastures help to prevent soil erosion.

_____

_____

_____

_____(2 marks)

b. A farmer established a grass pasture using the following information:

Seed size = 200 000 seeds/kg
Purity = 75%
Germination = 50%
If the expected plant population was 60 000 plants per hectare, calculate seed rate of the grass pasture. Show your working.

(3 marks)

7. Give **three** ways in which burning is important in pasture management.

_____

_____

_____

_____(3 marks)

8.

a. Differentiate between Macro and micro plant nutrients

_____

_____(1 mark)

b. Give **three** conditions that make plant nutrients to be essential.

_____

_____

_____

_____(3 marks)

9.

a. Explain **two** function of nitrogen.

_____

_____

_____

_____(2 marks)

b. State any **two** ways in which potassium can be depleted from the soil.

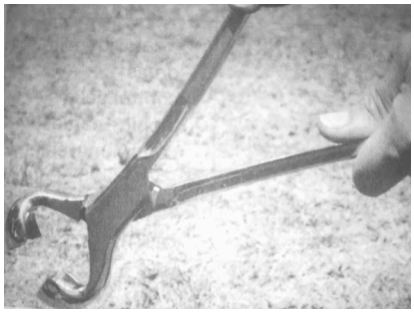
_____

_____

_____

_____(2 marks)

10. Below is a picture showing equipment used in livestock production.



a. Identify the equipment

_____(1 mark)

b. Explain how it can be used in livestock production.

_____

_____

_____

_____(2 marks)

c. State **two** importance of the practice carried out using the equipment.

_____

_____

_____

_____(2 marks)

Section B (40 marks)

Answer **all four** questions in this section. Your answers should be in essay form.

11. Explain any **five** ways HIV/AIDS impacts agricultural development.

_____(10 marks)

12. Explain any **five** limitations of mechanization.

(10 marks)

13. Explain any **five** importance of organic matter in crop production

_____(10 marks)

14. Describe **five** factors that affect demand of agricultural commodities.

_____(10 marks)

END OF QUESTION PAPER